Schema Width Bounds the Benefit of Editable Keys for Plurality-Based Functional-Dependency Checks

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November 10, 2026

Abstract

A plurality-based functional-dependency check accepts a target's dependent value when it is the unique plurality among rows that agree with the target on a determinant. We study the minimum number of context-cell edits that makes every listed check accept a protected, initially rejected target, and ask how much cheaper this becomes when determinant (key) cells may change as well as dependent cells. For a family F of determinants sharing one dependent attribute, let $w(F)$ be the size of a largest inclusion antichain among the determinants that contain a common key. We prove that the fixed-key optimum is at most $w(F)$ times the editable-key optimum, and that every family has a binary table attaining exactly this factor. Hence the two optima agree on every such table if and only if F is laminar. With at least $k+1$ dependent values the factor is also attained at editable-key optimum k, whereas for a binary dependent attribute and two overlapping determinants the optima differ by at most one edit. The upper bound uses a margin-capped exchange that replaces each key edit by at most one dependent-cell flip per inclusion chain. The classification concerns the listed checks, not logical FD equivalence, and laminarity does not make optimization easy: acceptance within a budget remains NP-complete for singleton determinants. For the reverse direction we give an exact formula. Exhaustive finite checks of the main theorems cover 81,893 rejected table/family pairs. On Beers, Flights and Rayyan, all nine check families of our own earlier benchmark experiment are laminar, which explains, post hoc, why the accepting gain it predicted from editable keys could not occur.

1 Introduction

Functional dependencies (FDs) describe agreement between attributes. A simple detector judges an observed dependent value by comparing it with the unique plurality among rows that share its determinant values. Plurality choices are a common primitive in FD-based cleaning. Equivalence-class repair assigns each class the value of least total modification cost [15], which under unit weights and 0/1 distance is a most frequent value. Horizon chooses repairs that preserve the most frequent FD-induced patterns [16]. Such a decision depends on context. Editing another row's dependent cell changes a vote; editing one of its key cells moves the row into or out of the target's group. An audit that trusts all key cells therefore answers a different question from one that allows them to be wrong.

We measure this as an exact decision-sensitivity problem. The target row is immutable, every changed context cell costs one, and the edited table need not satisfy any FD. Let ρ_{fixed} be the minimum cost of making all checks accept a rejected target when only dependent cells may change, and ρ_{full} the minimum when key cells may change too. Always $\rho_{\text{full}} \le \rho_{\text{fixed}}$. The question is how large the gap can be, and which schemas allow a gap at all.

A four-row example. Take rules $(a, b) \to y$ and $(a, c) \to y$ and the binary table in Table 1. Each target group contains the target and one rival, so both checks tie and reject. With fixed keys, $r_1.y$ and $r_2.y$ must both change: $\rho_{\text{fixed}} = 2$. Changing only $r_3.a$ to 0 moves r_3 into both groups at once, giving $\rho_{\text{full}} = 1$. The two determinants share the key a but neither contains the other. Such a family arises, for instance, when a city attribute is checked from both (zip, street) and (zip, phone prefix). Our main result shows that overlapping but non-nested determinants are the only source of a gap, and measures exactly how large it can become.

Row	a	b	c	y
Target (protected)	0	0	0	0
r_1	0	0	1	1
r_2	0	1	0	1
r_3	1	0	0	0

Table 1: Constructed illustration, not source data. One key edit replaces two dependent-cell edits.

Contributions.

1. For every listed family F and every initially rejected target we prove $\rho_{\text{full}} \le \rho_{\text{fixed}} \le w(F)\,\rho_{\text{full}}$, where $w(F)$ is the maximum, over keys j, of the inclusion width of the determinants containing j (Theorem 1). The proof converts any optimal editable-key witness into a fixed-key witness.

2. The factor is sharp for each family separately: some binary table has $\rho_{\text{full}} = 1$ and $\rho_{\text{fixed}} = w(F)$ (Theorem 2). Consequently the two optima coincide on every table with a rejected target exactly when F is laminar (Corollary 1). The factor is also attained at every optimum k when the dependent attribute has at least $k+1$ values, but not in general when it is binary (Proposition 1).

3. The equality is a statement about worst cases and edit spaces, not about tractability. Making a rejected target accept within a budget is NP-complete for binary singleton schemas, which are laminar (Theorem 4). For the opposite direction, rejection, an exact formula shows that keys help only through singleton groups (Theorem 3).

4. Exhaustive checks of raw edit spaces, run under protocols fixed in advance, agree with every finite prediction. A post hoc reanalysis of our own earlier benchmark experiment shows that its schemas were all laminar, so the accepting gain we had predicted was impossible; we report this design error rather than changing the rules after the fact.

The chain-partition step uses Dilworth's theorem [1]; nested-set dominance also appears in laminar election-manipulation algorithms [4]. We claim the specific exchange ratio, its sharpness for each family, and the laminar characterization, not these techniques. A targeted search found no identical statement (Section 7); this is not a historical-priority guarantee. A radius certifies stability of a decision under a stated edit model. It does not say which cells are wrong and is not evidence of better cleaning accuracy.

2 Model

Let $T = (t, r_1, \ldots, r_{n-1})$ be identified rows with $q \ge 1$ categorical key attributes and one dependent attribute y, the right-hand side (RHS) of every rule. Duplicate rows are allowed. The RHS domain

has $d \geq 2$ values; each key domain contains the target value and an alternative. By relabeling each column, assume the target is all zero. A nonempty family F of distinct nonempty subsets of $\{1, \ldots, q\}$ lists the checks $X \to y$, $X \in F$. Define

$$G_X(T) = \{r \in T : r[X] = t[X]\}, \qquad c_{Xv}(T) = |\{r \in G_X(T) : r_y = v\}|.$$

The target is accepted iff $c_{X0}(T) > c_{Xv}(T)$ for every $X \in F$ and $v \neq 0$. Ties reject, and a plurality need not be a majority when $d > 2$.

An edited table keeps the same rows and leaves t unchanged. Context cells may independently take any value in their domains. The cost is the cell Hamming distance $\delta(T, T')$. For a target whose decision is to be flipped, ρ_{full} is the minimum cost of a flip and ρ_{fixed} the minimum over edits that keep every key cell; an impossible flip has cost ∞. There are no insertions, deletions, unequal costs, validity constraints or requirements on other targets. These restrictions are used in the proofs.

For the accepting direction write $M(r) = \{j : r_j = 0\}$ for the zero-key mask of a context row and define its contribution to rule X and rival $v \neq 0$ as

$$a_{Xv}(r) = \mathbf{1}[X \subseteq M(r)]\big(\mathbf{1}[r_y = 0] - \mathbf{1}[r_y = v]\big).$$

The margin is $m_{Xv}(T) = 1 + \sum_{r \neq t} a_{Xv}(r)$, where the 1 is the protected target's vote. Acceptance means $m_{Xv} \geq 1$ for all X, v. For an initially rejected target both radii are finite: changing every nonzero context y to 0 accepts.

3 Schema width and the accepting gap

Throughout this section the target is initially rejected, and ρ_{full}, ρ_{fixed} are the costs of making it accept.

For a key j let $F_j = \{X \in F : j \in X\}$, partially ordered by inclusion. Let w_j be the size of a largest antichain in F_j (zero if F_j is empty) and

$$w(F) = \max_j w_j \ \geq 1.$$

F is *laminar* if any two intersecting determinants are nested. Each w_j is the width of a poset with at most $|F|$ elements. By Dilworth's theorem it equals a minimum chain partition, which reduces to bipartite matching [17], so $w(F)$ is computable in time polynomial in $|F|$ and q. Table 2 lists simple cases. The last row uses Sperner's theorem on the subsets of the remaining $q - 1$ keys.

Listed family F	$w(F)$	Laminar
Singletons $\{1\}, \ldots, \{q\}$	1	yes
A chain $\{1\} \subset \{1,2\} \subset \cdots \subset \{1, \ldots, q\}$	1	yes
$\{a,b\}, \{a,c\}$	2	no
All $\binom{q}{2}$ pairs, $q \geq 3$	$q-1$	no
All nonempty subsets	$\binom{q-1}{\lfloor (q-1)/2 \rfloor}$	no for $q \geq 3$

Table 2: Width of some determinant families.

Lemma 1 (Normalization)**.** *For every original context row r and destination u there is a destination c with $\delta(r,c) \leq \delta(r,u)$ and $a_{Xv}(c) \geq a_{Xv}(u)$ for all X, v, of one of two forms: (i) $c = r$ and $r_y \neq 0$; or (ii) $c_y = 0$, $M(c) \supseteq M(r)$, and every changed key of c is set to 0.*

Proof. Suppose $r_y \neq 0$ and $u_y = r_y$. If $u = r$, take form (i). Otherwise u changes a key, so $\delta(r, u) \geq 1$; take c with the original keys and $c_y = 0$, which costs one. Every contribution of c is nonnegative and every contribution of u nonpositive.

In all other cases let $c_y = 0$, set the keys in $M(r) \cup M(u)$ to 0, and keep the original value elsewhere. The key cells changed by c lie in $M(u) \setminus M(r)$ and are also changed by u. If $r_y = 0$ the y cost of c is zero; otherwise it is one and $u_y \neq r_y$ also costs one. If $u_y = 0$, every group containing u contains c. If $u_y \neq 0$, the contributions of u are nonpositive while those of c are nonnegative. Since margins are sums over rows, rows can be replaced independently without increasing cost or decreasing any margin. □

Thus some optimal editable-key witness uses only two row actions: keep a rival row, or set its y to 0 and add zero-key matches. A key edit never removes a rival from a group. Each row has at most $1 + 2^{q-|M(r)|}$ actions.

Theorem 1 (Exchange bound)**.** *For every listed family and every table whose target is initially rejected,*

$$\rho_{\text{full}} \leq \rho_{\text{fixed}} \leq w(F)\, \rho_{\text{full}}.$$

More precisely, a normalized accepting witness with R dependent-cell edits and K_j edits in key column j yields a fixed-key accepting witness of cost at most $R + \sum_j w_j K_j$.

Proof. The first inequality holds because fixed-key edits are a subset. For the second take an optimal editable-key witness, normalized by Lemma 1. By Dilworth's theorem [1, Theorem 1.1], partition each F_j into w_j inclusion chains.

Apply the R dependent-cell edits first and copy the result into two tables: H, which will execute the key edits, and L, which keeps the original keys and may flip further nonzero y values to 0. Maintain for all X, v

$$\min\big(m_{Xv}(L), 1\big) \;\geq\; \min\big(m_{Xv}(H), 1\big). \tag{1}$$

Initially equality holds. Consider the next key edit in H, in column j. By normalization it sets a key to 0 in a row whose y is already 0. The row joins exactly the groups of some determinants in F_j, the *affected* rules, and each of their margins rises by one. No other margin changes.

For each chain of F_j, look at its affected rules whose L-group still contains a context row with nonzero y. If there is none, do nothing. Otherwise let X be the largest such determinant and flip one nonzero-y row of $G_X(L)$ to 0. For any other such rule Y in the chain, $Y \subseteq X$, so $G_Y(L) \supseteq G_X(L)$ contains the flipped row; each of its margins rises by at least one (by two for the row's former value). An affected rule whose L-group has no nonzero-y context row already has $m_{Xv}(L) \geq 1$ from the target. A flip to 0 never lowers any margin.

Hence for each affected margin either $m_{Xv}(L) \geq 1$ already, or $m_{Xv}(L) \geq m_{Xv}(H)$ by (1) and L gains at least as much as H. Unaffected margins of H do not change. So (1) persists, using at most w_j flips for this key edit. Flipped rows had nonzero y, so they are distinct from each other and from the R edited rows. When all key edits are processed, H accepts; by (1) so does L, which never changed a key or the target. Its cost is at most $R + \sum_j w_j K_j \leq w(F)(R + \sum_j K_j) = w(F)\, \rho_{\text{full}}$. □

The cap at one in (1) is needed: a group with no rival left in L may have a smaller margin than in H yet already accepts.

Theorem 2 (Sharpness for every family)**.** *For every listed family F there is a binary table with an initially rejected target such that $\rho_{\text{full}} = 1$ and $\rho_{\text{fixed}} = w(F)$.*

Proof. Choose j with $w_j = w = w(F)$ and an antichain $A_1, \ldots, A_w$ in F_j. In the rows below, every key outside the stated zero mask is 1.

- For each i, a row r_i with $y = 1$ and zero mask exactly A_i.
- A row p with $y = 0$ and every key 0 except key j.
- For each $C \in F$ that contains none of the A_i, w support rows with $y = 0$ and zero mask exactly C.

The group of A_i is $\{t, r_i\}$: r_l with $l \neq i$ would need $A_i \subseteq A_l$; p fails key $j \in A_i$; a support row for C would need $A_i \subseteq C$. So each such check ties and rejects.

Every other check accepts. If C contains no A_i, its group has the target and w support rows with $y = 0$, against at most w rivals. If $A_i \subseteq C \neq A_i$ and a rival r_l were in G_C, then $A_i \subseteq C \subseteq A_l$, forcing $l = i$ and $C = A_i$; so G_C has no rival.

Changing p_j to 0 puts p into every group, making every critical group 2:1 and harming no check; as the target was rejected, $\rho_{\text{full}} = 1$. With fixed keys each critical group contains only the target and its own r_i, so the w distinct cells $r_i.y$ must all change to 0 (for $d > 2$ a third value still ties), and changing them suffices. Thus $\rho_{\text{fixed}} = w$. The table has at most $w + 1 + w|F|$ context rows. □

Corollary 1 (Laminar characterization)**.** *The following are equivalent: (1) $\rho_{\text{fixed}} = \rho_{\text{full}}$ for every table with an initially rejected target; (2) $w(F) = 1$; (3) F is laminar.*

Proof. Two intersecting incomparable determinants share some key j and form a two-element antichain in F_j, and conversely. This gives (2)⇔(3). Theorems 1 and 2 give (1)⇔(2). □

The restriction to rejected targets is necessary. For the single rule $x \to y$, target $(0, 0)$ and one context row $(1, 1)$, the target is accepted; one key edit makes it reject, but with fixed keys its group stays $\{t\}$ and $\rho_{\text{fixed}} = \infty$ (Theorem 3), although $\{x\}$ is laminar.

Larger optima. Theorem 2 attains the factor with $\rho_{\text{full}} = 1$. Whether it is attained at larger optima depends on the number of RHS values.

Proposition 1 (Larger optima)**.** *(a) Let $k \geq 1$ and $d \geq k + 1$. For every listed family F there is a table with binary keys and an initially rejected target such that $\rho_{\text{full}} = k$ and $\rho_{\text{fixed}} = w(F)\, k$.*

(b) Let $d = 2$ and $F = \{\{a, b\}, \{a, c\}\}$. Every table with an initially rejected target has $\rho_{\text{fixed}} \leq \rho_{\text{full}} + 1$. In particular $\rho_{\text{fixed}} = 2\, \rho_{\text{full}}$ forces $\rho_{\text{full}} = 1$.

Proof. (a) Choose j and the antichain $A_1, \ldots, A_w$ as in Theorem 2; every key outside a stated zero mask is 1.

- For each i and each value $v \in \{1, \ldots, k\}$, k rows with $y = v$ and zero mask exactly A_i.
- k rows with $y = 0$ and every key 0 except key j.
- For each $C \in F$ that contains none of the A_i, wk support rows with $y = 0$ and zero mask exactly C.

By the argument of Theorem 2, the group of A_i is the target together with its own k^2 rivals: one zero against k votes for each of k values, so the check rejects. A group of C containing no A_i has at least $1 + wk$ zeros and at most wk rows of any one rival value, and a group of $C \supseteq A_i$ with $C \neq A_i$ has no rival; these checks accept.

Setting key j of the k zero rows to 0 puts them into every group. Each critical group then has $k+1$ zeros against k votes per rival value, and no check is harmed, so $\rho_{\text{full}} \le k$. With fixed keys, suppose fewer than k dependent cells change among the rivals of one critical group. Some rival value keeps all k of its rows, while the zeros number at most $1+(k-1)=k$, so the check still rejects. Hence each critical group needs k changes among its own rows, and these row sets are disjoint: $\rho_{\text{fixed}} \ge wk$. Changing one row of each rival value to 0 in each critical group gives $k+1$ zeros against $k-1$, so $\rho_{\text{fixed}} = wk$. Theorem 1 now gives $\rho_{\text{full}} \ge k$.

(b) Take an optimal editable-key witness, normalized by Lemma 1, with R dependent-cell edits and K key edits, so $R+K=\rho_{\text{full}}$. Apply the R dependent-cell edits, which keep every key, and call the result T_1. Since $d=2$ each rule X has one margin $m_X = m_{X1}(T_1)$; let $n_X = \max(0, 1-m_X)$. By normalization the key edits only move rows with $y=0$ into groups. Each such row raises a margin by one and carries at least one of the K key edits, so acceptance of the witness gives $n_X \le K$. With fixed keys, instead flip $\lceil n_X/2 \rceil$ rows with $y=1$ in $G_X(T_1)$ to 0, for each rule X. Each flip raises m_X by two and lowers no margin. There are enough such rows: if $n_X > 0$, the group has $c_{X1} = c_{X0} - m_X \ge 1 - m_X = n_X$ of them. The result accepts, so

$$\rho_{\text{fixed}} \;\le\; R + \sum_{X \in F} \lceil n_X/2 \rceil \;\le\; R + |F|\,\lceil K/2 \rceil,$$

which for two rules is at most $R+K+1 = \rho_{\text{full}} + 1$. If $\rho_{\text{full}} \ge 2$ then $\rho_{\text{full}} + 1 < 2\,\rho_{\text{full}}$. □

For $k=1$ the table of (a) is that of Theorem 2. With k distinct rival values a flip to 0 lowers only one of k tied rivals, whereas one flip raises a binary margin by two. The displayed inequality in (b) holds for every family with binary RHS, but improves on Theorem 1 only when $|F|$ is small. The case $d \le k$ is open in general.

Remark 1 (What the result does and does not say). The factor $w(F)$ is the largest *possible* gain; nonlaminar schemas have many tables with no gain. For laminar schemas the editable-key search can be replaced by the fixed-key search, whose only variables are context dependent cells. That reduction removes key variables but not hardness (Theorem 4). For general F, any exact fixed-key value is a $w(F)$-approximation of ρ_{full}. The exchange needs an optimal editable-key witness as input, so it is not itself an approximation algorithm.

Remark 2 (Listed checks versus logical FDs). The result concerns the listed plurality checks, not the FDs they express: logically equivalent families can behave differently. Add rule $\{a\}$ to Table 1 together with two rows $(a,b,c,y) = (0,1,1,0)$. The $\{a\}$ group has three zeros and two ones and accepts, while the two original groups still tie, so $F = \{\{a\},\{a,b\},\{a,c\}\}$ still has $\rho_{\text{fixed}} = 2$, $\rho_{\text{full}} = 1$ and $w(F) = 2$. As FDs, $a \to y$ implies $ab \to y$ and $ac \to y$, so F is equivalent to $\{\{a\}\}$; yet the single check $\{a\}$ has width one and already accepts this table.

4 Complementary results

4.1 Making an accepted target reject

Assume the target is accepted. Write $\alpha_X = c_{X0}(T)$, $\beta_X = \max_{v \ne 0} c_{Xv}(T)$ and $h_X(r) = |\{j \in X : r_j \ne 0\}|$.

Theorem 3 (Exact rejection radius)**.** *Let* $\kappa_X = \lceil (\alpha_X - \beta_X)/2 \rceil$ *if* $|G_X(T)| \ge 2$*, and* $\kappa_X = \min_{r \ne t}\big(h_X(r) + \mathbf{1}[r_y = 0]\big)$ *if* $G_X(T) = \{t\}$*, with an empty minimum equal to* ∞*. Then* $\rho_{\text{full}} = \min_X \kappa_X$*. With fixed keys,* ρ_{fixed} *is the minimum over nonsingleton groups only.*

Proof. In a nonsingleton group a key edit changes a target/rival margin by at most one and an RHS edit by at most two, so rejection costs at least $\lceil(\alpha_X - \beta_X)/2\rceil$. Changing that many zero-valued context RHS cells in the group to a strongest rival attains it. There are enough, because acceptance with a nonsingleton group gives $\alpha_X \geq 2$ and $\lceil(\alpha_X - \beta_X)/2\rceil \leq \alpha_X - 1$. A singleton group must receive a context row with nonzero y, which pays for its $h_X(r)$ nonmatching keys and, if needed, its RHS; editing just that row creates a tie. Global rejection is a disjunction over rules, so the minimum over rules is exact. With fixed keys a singleton group cannot change. □

Thus editable keys strictly lower the rejection radius iff the cheapest singleton import costs less than every nonsingleton margin cost. The two decision directions behave differently: rejection depends on singleton groups, acceptance on the width of the family.

4.2 Complexity of acceptance

Theorem 4 (Binary singleton acceptance is NP-complete)**.** *Deciding whether a rejected target can be accepted within budget B is NP-complete in combined complexity, even with binary columns, singleton determinants and initially tied target groups, with fixed or editable keys.*

Proof. Reduce from Set Cover [6] with universe $U = \{1, \ldots, q\}$ and sets $S_1, \ldots, S_\ell$ covering U. Let $c_g = |\{i : g \in S_i\}|$ and τ the minimum cover size. Use binary keys $x_1, \ldots, x_q$, rules $x_g \to y$ and an all-zero target. For each S_i add a row with $y = 1$ and $x_g = 0$ exactly for $g \in S_i$; for each g add $c_g - 1$ rows with $y = 0$, $x_g = 0$ and all other keys 1. Every group has c_g zeros and c_g ones. Flipping the y cells of a cover's rows accepts, so both optima are at most τ.

Conversely, take an accepting table of cost K. For each changed key cell in column x_g choose an input set containing g; for each set row whose final y is 0 choose its set. At most K sets are chosen. If column x_g has no key edit, its group membership is unchanged and, starting from a tie, some original $y = 1$ member, a set row containing g, must end at 0. So the chosen sets cover U and $\tau \leq K$. Both optima equal τ. A destination table is a polynomial certificate. □

Singleton families are laminar, consistent with both optima being τ. For fixed q and d the problem is polynomial in n. Margins are sums of per-row contributions, each margin lies in $[-n, n+1]$, and by Lemma 1 each row has at most $1 + 2^q$ useful actions. A dynamic program over rows that stores the least cost for each vector of the $|F|(d-1)$ margins therefore runs in time $O\big(n\, 2^q (2n+2)^{|F|(d-1)}\big)$. With fixed keys the only useful action is setting y to 0. The finite check of Proposition 1 (Section 5) uses a program of this kind over all row destinations, without the normalization; its running time was not studied.

4.3 Decision distance is not repair distance

Let $\mathrm{rep}(T)$ be the minimum context-cell cost making the whole table satisfy every FD, still protecting t.

Proposition 2. *Even for one rule $x \to y$ with binary y, both $\rho_{\mathrm{full}}/\mathrm{rep}(T)$ and $\mathrm{rep}(T)/\rho_{\mathrm{full}}$ are unbounded over instances where both are positive and finite.*

Proof. With target $(0, 0)$, $2h$ rows $(0, 0)$ and one row $(0, 1)$, one edit repairs the table but Theorem 3 gives $\rho_{\mathrm{full}} = h$. With target $(0, 0)$ and pairs $(i, 0), (i, 1)$ for $i = 1, \ldots, k$, each disjoint conflicting pair needs its own edit, so $\mathrm{rep}(T) = k$, while moving $(1, 1)$ into the singleton target group creates a tie at cost one. □

5 Finite checks

Each check below followed a protocol fixed in the project repository before its implementation, and the implementation was fixed before execution. Counts reuse tables and overlap across edit models; they measure coverage of the checks, not probabilities, and they do not replace the proofs.

Exhaustive check of Theorems 1–2 (E013). Three grids use an all-zero target, every multiset of context rows of the given size, and every family of distinct nonempty determinants: binary keys and RHS with $q \le 3$ and up to three context rows; binary keys, ternary RHS, $q = 3$, up to two rows; ternary keys and RHS, $q = 2$, up to two rows. Two raw oracles enumerate all destination tables (editable keys) and all RHS-only destinations (fixed keys), counting votes directly; neither uses the theorem. Width is computed by exhaustive antichains and checked against an independent minimum chain cover. Every optimal editable-key witness is converted by the procedure of Theorem 1, checking cost, unchanged keys, acceptance and the invariant (1) after each key edit. Table 3 gives the totals. All predictions hold; all 62 strict gains occur in the binary $q = 3$, three-row grid. The sharp construction is checked for every family with $q \le 3$ and four larger families with widths up to six; for these, the lower bound is checked by the isolated-group certificate rather than raw enumeration.

Quantity	Count
Source tables	1,895
Rejected table/family pairs checked	81,893
of which laminar (equality predicted and observed)	39,379
Pairs with $\rho_{\text{fixed}} > \rho_{\text{full}}$	62
Raw destinations, editable keys / fixed keys	3,892,339 / 14,687
Key edits converted / replacement flips	11,944 / 12,014
Single-row normalization checks	1,641
Sharp constructions checked	139

Table 3: E013 totals (protocol `e25655d`, implementation `b93695a`).

Check of Proposition 1 (E014). The construction of part (a) was built for nine cases: $\{\{a,b\},\{a,c\}\}$ with $(k,d) = (1,2),(2,3),(2,4),(3,4)$, and five further families on three or four keys with $k = 2$, $d = 3$ and widths two and three, including all seven nonempty subsets of three keys. In every case $(\rho_{\text{full}}, \rho_{\text{fixed}}) = (k, wk)$. Both radii come from a dynamic program over all row destinations, which uses neither the normalization nor the exchange. A binary control with $k = 2$ has $\rho_{\text{full}} = \rho_{\text{fixed}} = 2$, so the condition $d \ge k + 1$ cannot simply be dropped. In all ten tables every set of fewer cells than the editable-key radius, and of fewer dependent cells than the fixed-key radius, was also enumerated with direct vote counting: none of 5,594,099 edited tables accepts. For part (b), all 74,612 multisets of one to six context rows over three binary keys and a binary RHS were solved for $\{\{a,b\},\{a,c\}\}$. Of the 34,393 tables with a rejected target, 332 have a strict gap: 309 with $(\rho_{\text{full}}, \rho_{\text{fixed}}) = (1,2)$ and 23 with $(2,3)$. So $\rho_{\text{fixed}} \le \rho_{\text{full}} + 1$ throughout, and the slack of one is attained at $\rho_{\text{full}} = 2$. On the 968 tables with at most three rows the dynamic program agrees with raw enumeration of all 3,377,408 destination tables (protocol `6c8827a`, implementation `d98e454`, one run).

Exhaustive check of the reduction (E012). Every coverable multiset of one to three subsets of universes of size one to three (111 instances) is checked by raw binary enumeration: all 222 radii

and 147,523 destinations across both edit models agree with independent minimum covers, and a cover is extracted from all 25,490 accepting destinations.

Earlier checks (E008–E010). E008 compares canonical-action programs with unrestricted raw edits on 5,107 tables and six families (30,642 radii agree). E009 checks Theorem 3 on all 18,779 accepted cases of that domain. E010 compares a specialized and a generic MILP on 72 constructed cases with up to six keys and 128 rows; certified optima agree, and seven generic dense-accepted cases time out (two repetitions on one machine, not representative).

6 Benchmark study and its post hoc reinterpretation

Before the width theorem was found, we ran a benchmark experiment (E011) that reused Beers, Flights and Rayyan from the Raha repository [21] at revision `7be1334b`. The data had been inspected earlier, so this is exploratory reuse, not a blind holdout. Nine tasks use brewery_id to check brewery_name, city and state; flight to check four time columns; Rayyan's `jounral_abbreviation` (original spelling) to check journal_issn; and abbreviation and ISSN as two separate determinants for journal_title. Every target uses its full dirty table as context; empty strings are ordinary categories.

Of 18,734 target cells, 13,762 were accepted and 4,972 rejected. For accepted targets every rejection radius was computed. Editable keys turned 1,133 fixed-key infinite radii finite and lowered one finite radius (Rayyan journal_title row 644, from two to one), always through singleton groups, as Theorem 3 predicts. Of the accepted targets, 1,816 differ from the reference value and 1,493 of these have radius at least two: a stable acceptance can still be wrong.

For rejected targets, a hash-ordered selection of up to four per task gave 32 targets. Each editable-key and fixed-key MILP ran once with a five-second limit. All 32 fixed solves and 30 editable-key solves were optimal, and every exact pair had equal cost. The declared prediction of at least one strict accepting gain in the journal_title task was therefore not supported. Two editable-key models, rows 509 and 146 (one-based), timed out without bounds; their fixed optima are 104.

Post hoc reanalysis. After E013, we checked the archived E011 schemas. Every one is a singleton family, or two disjoint singletons for journal_title, hence laminar. By Corollary 1 all 4,972 rejected targets have equal fixed and editable-key accepting costs; the predicted gain could not occur under this edit model. Combining equality with the archived certified fixed optima gives editable-key optimum 104 for rows 509 and 146. This deduction rests on the proof and archived certificates. The solver records remain timeouts, the source CSV cache was unavailable, and no solver was rerun. The reanalysis was written after seeing the results and is labeled exploratory. Testing accepting gains on real data would require a nonlaminar check family justified and fixed before outcomes are seen.

7 Related work

Chains and laminar interventions. Dilworth's theorem [1] equates width with the minimum chain partition; we use it only for that step. In election manipulation under media influence, strategies sway voter sets. The slides of Miclăuș, Mureșan and Popa [4] credit Tao et al. [3] with a dynamic program for laminar influence sets, and show prefix dominance on chains of nested influence sets and chain-decomposition parameterized algorithms. We inspected the 2026 slides

and abstract only; neither full paper was obtained. That work optimizes within one action space, uses co-winner ties, and its laminar family consists of voter sets. Here laminarity is over column sets; the induced row groups all contain the target and so are never disjoint. We compare two edit spaces and give the exact worst-case ratio for each schema. Gerrymandering [5] studies interventions that move voters between districts, so membership-changing plurality interventions are established; the inspected problem definitions aggregate district winners and do not compare edit spaces.

Repair bounds depending on the schema. Livshits, Kimelfeld and Roy [2] (Proposition 4.4, Corollary 4.5) convert between optimal subset and update repairs of FD sets with a factor equal to the minimum lhs cover mlc, and use it for approximation. Their objective restores consistency of the whole table; ours flips one protected decision without requiring consistency (Proposition 2). The parameters differ: $\{\{a, b\}, \{a, c\}\}$ has mlc $= 1$ but $w = 2$, and q singletons have mlc $= q$ but $w = 1$.

Resilience, bribery and control. Resilience [18] is the minimum number of tuple deletions that makes a Boolean conjunctive query false, and is classified by query structure. Our radius is likewise a minimum intervention flipping a Boolean outcome, but it edits cells, concerns a plurality decision rather than a query, and our classification is of the gap between two edit spaces rather than of tractability. In social choice, dependent-cell edits resemble bribery that changes votes [19]. Key edits resemble control by adding or deleting voters [20], with one row voting in every check whose group it matches. Unweighted plurality bribery in a single election is polynomial [19]. The overlap of target groups across checks is what Theorem 4 exploits.

Causality, vote operations and lobbying. Cell-based causality for repairs [8], repair-based causes for query answers [9] and view-conditioned causality [10] study minimum interventions on cells; general vote-operation frameworks [11] optimize constructive and destructive goals. Set Cover incidence constructions are standard in multi-referendum lobbying [7]; our hardness proof credits that technique and adds the argument for arbitrary key edits. FD repair with trusted cells [12], certified nearest-neighbor robustness over repairs [13], and the distinction between dependency strength and detection value [14] are further context; like the latter, our radius is not an accuracy measure.

8 Limitations

The proofs need a common dependent attribute, unit and independent cell costs, an immutable target with the accepting value, and ties that reject. Different dependents, weighted costs, validity constraints, insertions or deletions, or co-winner semantics fall outside the stated results; Theorem 2 in particular uses ties. The width bound is worst case and says nothing about how often gaps occur. Proposition 1(b) treats one two-rule family only. Laminarity does not give a fast algorithm. The finite checks are small, the source study reuses inspected data, its rejected sample is small, and its post hoc reanalysis is not a confirmatory test. All proofs and checks were produced and reviewed inside one AI-assisted project; no independent human review has taken place. The literature comparison is targeted, not systematic.

9 Reproducibility and provenance

Commit identifiers refer to the public project repository `https://github.com/minggu151623/invariant-audit`. E013 protocol and implementation commits are `e25655d` and `b93695a`; the proof and the E011 reanalysis script were committed at `557f60d` before the script ran. E013 used Python 3.14.5 with the standard library only, and E014 (`6c8827a/d98e454`, run recorded at `27ca9b2`) Python 3.13.5 likewise. E012 uses `9f14171/65ad518`, E008 `efc1365/14fe214`, E009 `96962be/05a2b07`, E010 `fbd04cb/2212b58` and E011 `86dad7d/fd341d4`; E009–E011 used Python 3.13.5, NumPy 2.1.3 and SciPy 1.17.1 (HiGHS). Solver optima required a checked integer witness and bound agreement within 10^{-7}; timeouts stay censored. The supplement is the file `paper/context-edit-stability-r4/supplement.zip` in the repository. It contains the E013 protocol, code, summary, compressed case records, proof and a read-only replay that rechecks every stored witness and can repeat the frozen raw enumeration, and the E014 protocol, code and results, which the E014 script can recompute. Generative AI was used substantially for ideation, proofs, code and writing.

10 Conclusion

For plurality-based FD checks with a protected target, the benefit of allowing key edits when accepting a rejected target is governed by one schema parameter: the largest inclusion antichain among determinants sharing a key. The fixed-key optimum is within this factor of the editable-key optimum, the factor is attained for every schema (at every optimum k when $d > k$), and the optima always agree exactly for laminar schemas. This explains a negative empirical outcome and separates the question of when key errors matter from the question of how hard the optimization is.

Several questions remain open.

- *Small RHS domains.* What is the largest ratio $\rho_{\text{fixed}}/\rho_{\text{full}}$ at $\rho_{\text{full}} = k$ when $d \leq k$? Proposition 1 settles $d \geq k + 1$, and $d = 2$ only for two overlapping determinants.
- *Approximation.* We have not studied approximation algorithms for either radius; Theorem 1 only transfers a fixed-key approximation to ρ_{full} at a further loss of $w(F)$.
- *Extensions.* How do weighted costs, different dependent attributes, or co-winner ties change the classification?

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